



Question Set

Question 1

Expected exposure is the expected positive mark-to-market of a derivative at a future time. Given $EE(0)=0.0$, $EE(1)=3.2$, $EE(2)=4.1$, $EE(3)=2.0$ million, approximate exposure-years using the trapezoidal rule.

Question 2

Approximate the discounted exposure integral

$$\int_0^3 e^{-0.04t} EE(t) dt$$

using the same expected-exposure values from Question 1 and the trapezoidal rule.

Question 3

Use Simpson's 1/3 rule to approximate

$$\int_0^2 e^{-0.05t} dt$$

and compare with the exact value.

Question 4

Why does the Gaussian normal CDF require numerical integration or approximation? State the integral.

Question 5

Derive the change of variables used to compute

$$\mathbb{E}[(S - K)^+]$$

when

$$S$$

is lognormal.

Question 6

A desk estimates

$$\mathbb{E}[L \mid L > q]$$

by integrating the tail of a loss density. Explain why grid placement matters.

Question 7

Use the midpoint rule with four intervals to approximate

$$\int_0^1 x^2 dx$$

Include the error direction.

Question 8

Explain the difference between Monte Carlo integration and deterministic quadrature in pricing.

Question 9

Show that Simpson's rule is exact for any quadratic polynomial on two equal subintervals.

Question 10

A CVA approximation uses $\sum_i DF_i EE_i PD_i$. Explain this as a numerical integration formula.

Question 11

What convergence check would you run before trusting a numerical integral in a risk report?

Question 12

Why can interpolation dominate integration error in exposure calculations?